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	CODE :	DRC 1133
	ASSESSMENT :	Assignment 2

PROJECT TITLE:

“rm.” ARG Escape Room

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1.0 Project Overview

The proposed website, "**rm.**", is a web-based Alternate Reality Game (ARG) that challenges users to solve complex cryptography and hardware-related puzzles. The application demonstrates high proficiency in client-side scripting by utilizing **JavaScript** to manage game states, validate user input, and handle core mathematical logic.

To enhance security and simulate a realistic "Secure Terminal" environment, the project implements **SHA-256 hashing** for access key verification and utilizes **LocalStorage** for persistent user authentication. The project consists of four interconnected HTML pages, ensuring a complete and immersive user journey from account creation to interactive gameplay:

Page	File Name	Description
Registration	register.html	Secure entry point featuring real-time form validation and Background Music integration.
Login	login.html	Authentication gate using LocalStorage. Includes a Music Toggle to manage ambient audio.
The Room	rm.html	Core gameplay interface with an interactive SVG environment. Features a Hint Icon hotspot and triggers for the Core Calculation sequence.
The Hint	hint.html	Technical manual providing frequency data and scientific notation constants required for the mathematical escape puzzle.

2.0 Static vs. Dynamic Justification

While a static web application provides fixed content that remains the same for every user (such as a simple "About" page), "rm." is a dynamic web-based application. It utilizes client-side scripting to change content and behavior based on user interaction without needing to reload the page.

Specific Examples from "rm.":

- **Dynamic State Management:** The "The Room" interface uses JavaScript to update the DOM in real-time, such as revealing hidden clues or changing the terminal display when a puzzle is solved.
- **Persistent Data:** By using LocalStorage, the application dynamically retrieves and validates the user's "Agent ID," providing a personalized and stateful experience that a static site cannot offer.

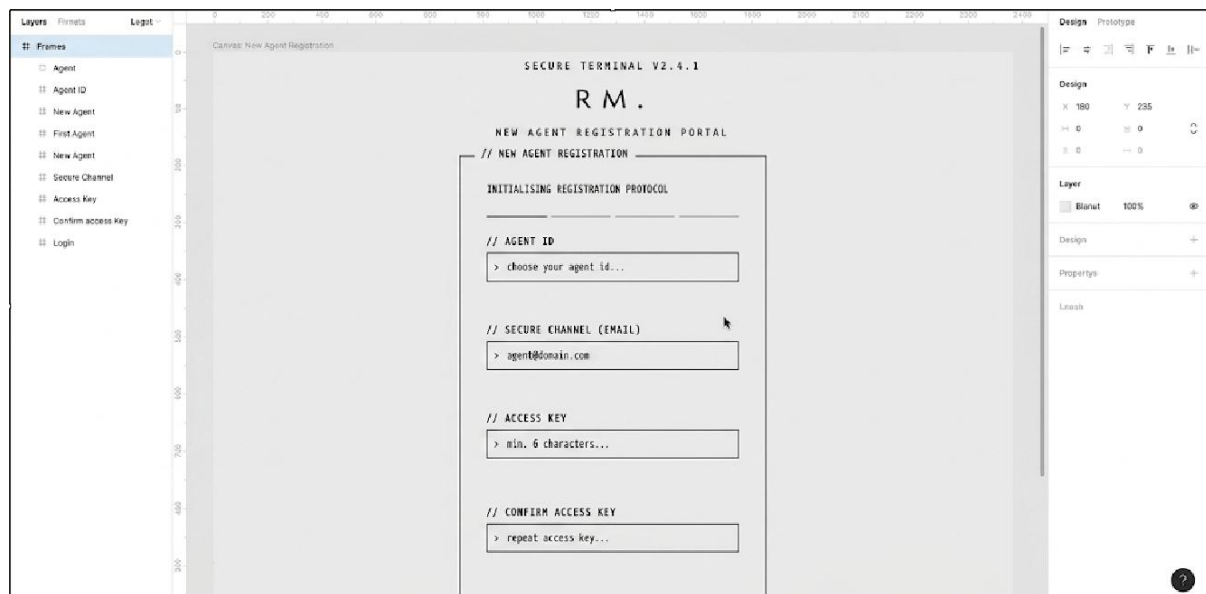
3.0 Wireframes

The following wireframes illustrate the layout and structure planned for each page of the 'rm.' ARG Escape Room website.

Page 1: Registration Portal (register.html)

Elaboration: Input Constraints & Feedback

- **Design Focus:** The wireframe includes specific placeholders like "// AGENT ID" and "// ACCESS KEY" to establish a command-line interface (CLI) aesthetic.
- **Logic:** I have designed the input fields to trigger real-time JavaScript validation; for example, the Access Key field specifies "min. 8 characters". This provides immediate UI feedback to the user, ensuring they meet security requirements before attempting to submit the form.



Page 2: Terminal Access Point (login.html)

Elaboration: Authentication Flow

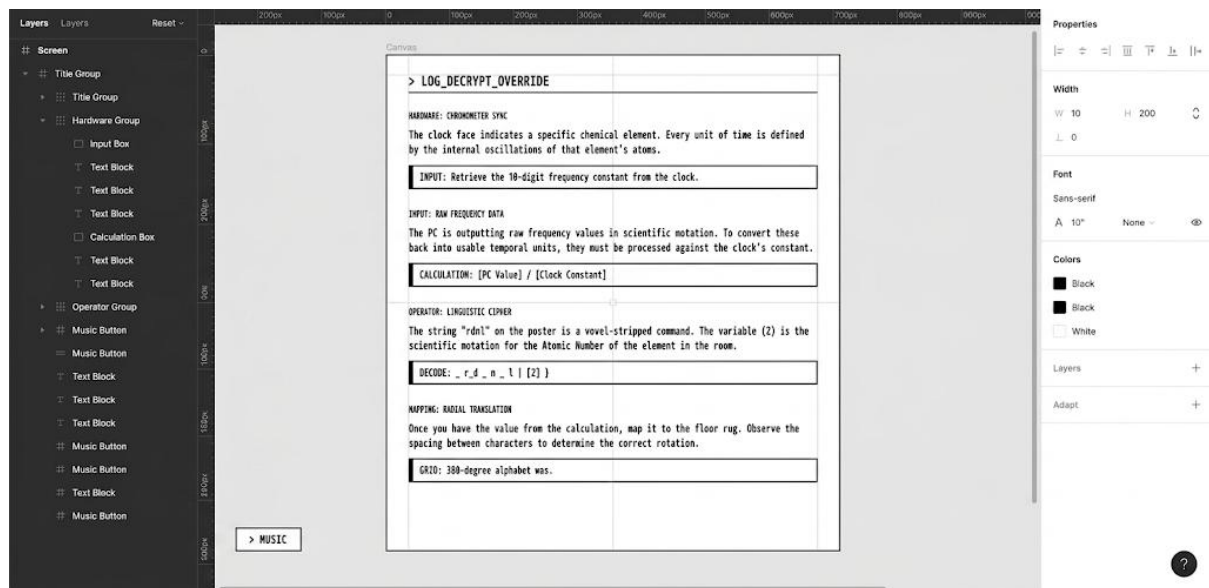
- **Design Focus:** The "AUTHORIZE" button is the primary Call-to-Action (CTA), centrally placed for high visual hierarchy.
- **Logic:** This wireframe represents the bridge between user input and localStorage. The design ensures that if a user has not registered, the system prevents progression, simulating a secure environment where only verified "Agent IDs" can initiate a decryption session.



Page 3: The Hint / Documentation (hint.html)

Elaboration: Information Hierarchy & Lore Integration

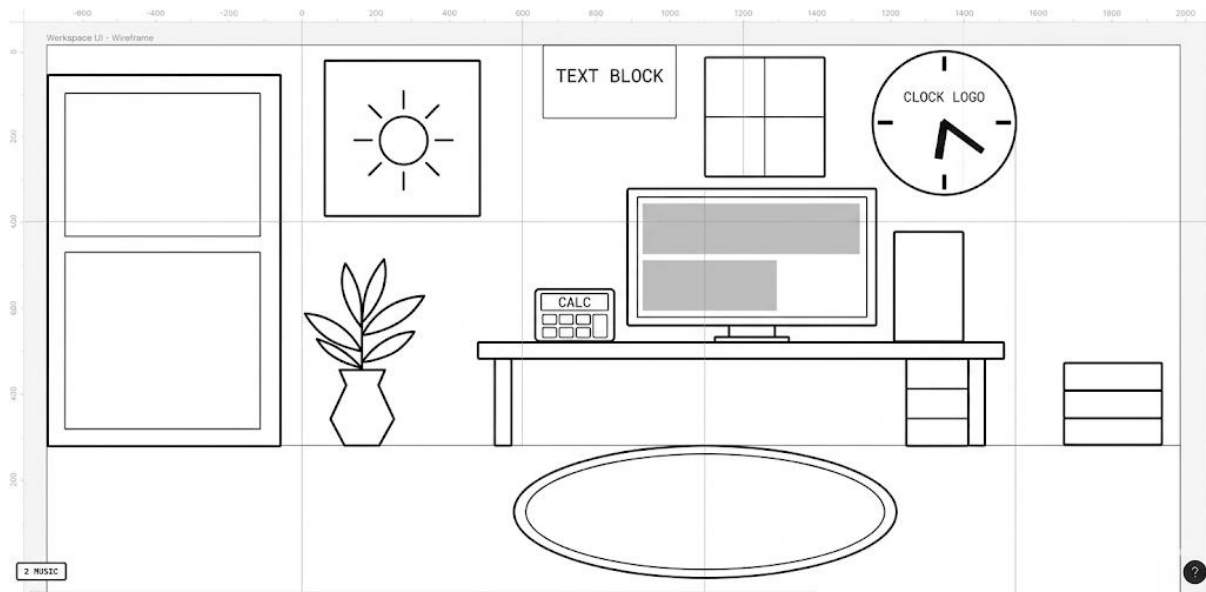
- **Design Focus:** This page is styled as a "Leaked Technical Manual" to provide the mathematical constants needed for the escape puzzle.
- **Logic:** To assist the user's cognitive load, I have organized the "Core Calculation" logic into distinct blocks (Input, Constant, Calculation, and Operator). This structured layout ensures the player can easily find the Cesium frequency constant (9,192,631,770 Hz) and the scientific notation variables required to solve the final terminal command.



Page 4: The Room (rm.html)

Elaboration: Interactive SVG Hotspots

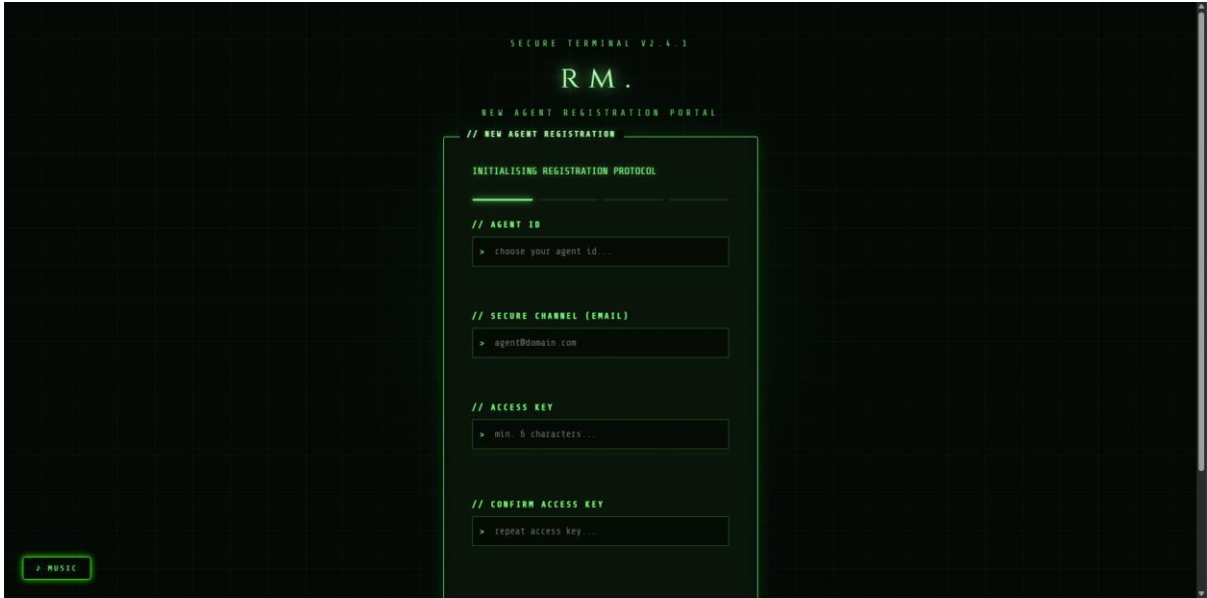
- **Design Focus:** The layout uses a custom SVG room where every object—like the clock, the PC, and the floor rug—is a functional hotspot.
- **Logic:** I have integrated a **Hint Icon (?)** as a persistent UI element. When clicked, it triggers a modal window via DOM manipulation, allowing the player to access technical data without leaving the room, which maintains immersion and supports "1-click" access to key features.



4.0 Website Screenshots & Descriptions

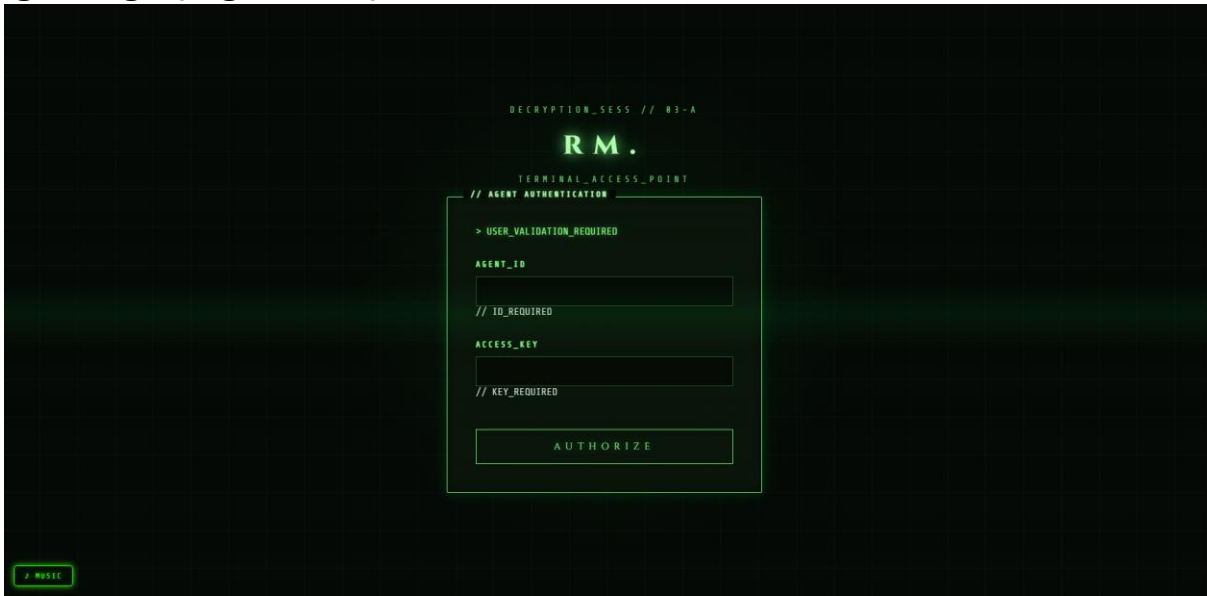
The following section describes each page of the "rm." ARG.

4.1 Register Page (register.html)



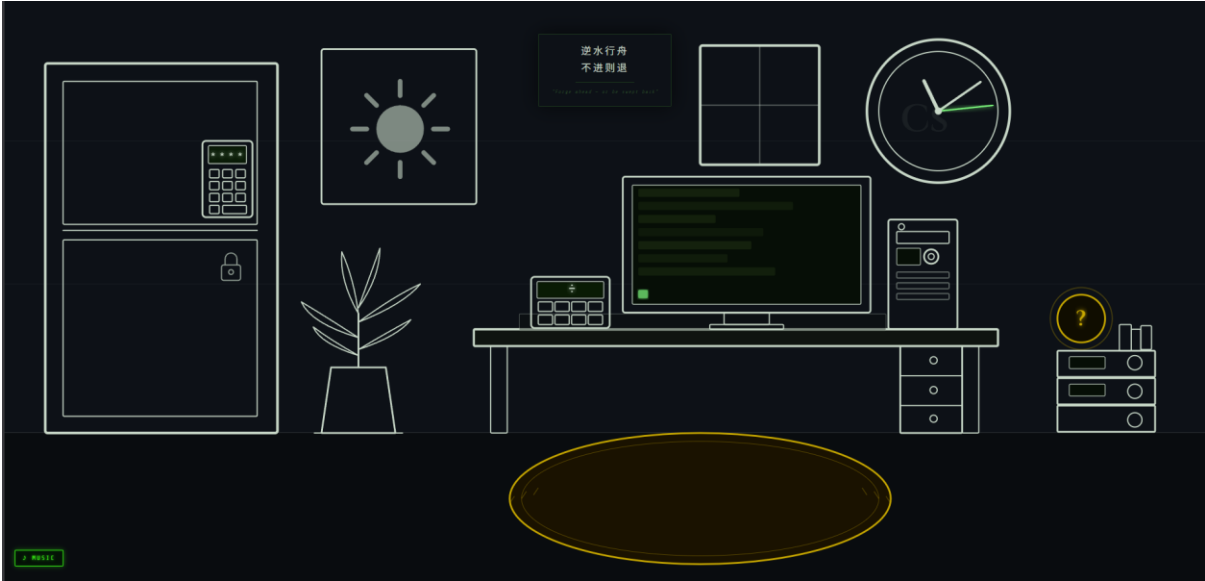
Purpose	To onboard new "operatives" into the game.
Key Features	Uses a terminal-style aesthetic. The JavaScript logic verifies credentials against localStorage and includes an interactive 🎵 MUSIC button that allows the user to play or pause the background track.

4.2 Login Page (login.html)



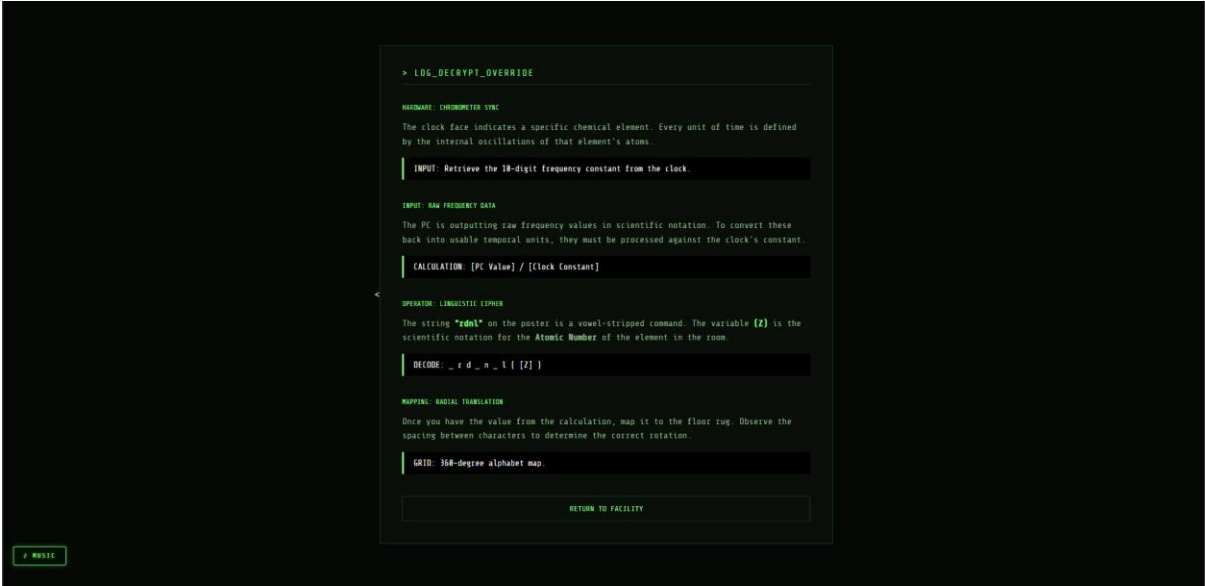
Purpose	To authenticate existing users
Key Features	Uses a terminal-style aesthetic. The JavaScript logic checks the input against the values stored in LocalStorage to verify credentials and maintain the user's session state.

4.3 The Room (rm.html)



Purpose	The primary gameplay screen where the user must find the "key."
Key Features	Features a custom SVG room with clickable hotspots. A new Hint Icon has been integrated as an interactive hotspot that reveals hidden puzzle data via a modal window. This page uses heavy DOM manipulation to manage the game state and sound effects.

4.4 The Hint/Documentation (hint.html)



Purpose	To provide the necessary lore and data to solve the final puzzle.
Key Features	This page now serves as the primary source for the Core Calculation logic. It contains "leaked" technical data showing frequency constants (9,192,631,770 Hz) and scientific notation values that the player must process using the terminal's calculation engine to progress.

4.0 References

[1] MDN Web Docs. (2024). *Using the Web Storage API*.

[2] Google Fonts. (2024). *Share Tech Mono (Cyberpunk Typography)*. 3. W3Schools. (2024). *How to Create an SVG Image Map*.

[3] Background Music (Register/Login):

<https://www.youtube.com/embed/rzgR40IceRM?si=KxnZUWtr781NgTOU>

[4] Background Music (Hint/RM Page):

<https://www.youtube.com/embed/Dvwaq1rwioQ?si=3zxPYjKrDqpPXkfT>

Name:

Appendix

Rubric : Part A

CO1: Distinguish the fundamentals of dynamics web-based applications (C2, PLO1)

SCORE						
Components	5 (Excellent)	4 (Good)	3 (Fair)	2 (Poor)	Weightage	Marks
Justification of Question 1	Provides a well-structured and clear justification, accurately distinguishing between static and dynamic web applications. Explanation is logically sound, supported with a relevant example from the proposed website.	Provides a justification that correctly differentiates static and dynamic web applications, with a somewhat relevant example from the proposed website, but lacking depth in explanation.	Justification is vague, partially correct, or lacks a relevant example to the proposed website. Limited understanding of static vs. dynamic web applications.	Justification is unclear, incorrect, or missing, with no attempt to differentiate static and dynamic web applications.	*1	/5
Instruction (Report Quality)	The report is complete, well-organized, and includes all required elements: table of contents, wireframes, screenshots with descriptions, references, and rubrics. The content is clearly written and effectively explains the website development process.	The report is mostly complete with minor missing elements or inconsistencies. Content is understandable with some areas needing clarity or more detail.	The report is partially complete with several missing elements. The content lacks organization or clarity in explaining the website development.	The report is incomplete, poorly organized, and fails to explain the website development process adequately.	*2	/10
Title (Usage of Headings)			The title accurately reflects the content and purpose of the web-based application, providing clear insight into its dynamics.	The title is unclear or irrelevant to the content and purpose of the webbased application.	*0.4	/2
Layouts	The layout of the web-based application is well-structured, with clear organization and visual hierarchy, facilitating ease of use and navigation.	The layout is generally well-organized, but there may be minor inconsistencies or issues with visual hierarchy.	The layout is somewhat cluttered or disorganized, making it challenging for users to navigate or locate specific components.	The layout is somewhat cluttered or disorganized, making it challenging for users to navigate or locate specific components.	*1	/5
Links			All links within the web-based application are functional and lead to relevant destinations, enhancing user interaction and exploration	There is at least three (3) link. Many links are broken or lead to irrelevant destinations, severely impacting the functionality and usability of the web-based application.	*0.4	/2
Images			Images are strategically used throughout the web-based application to enhance visual appeal, reinforce content, and engage users effectively.	The image(s) is either absent or poorly integrated into the web-based application, failing to enhance user experience or support content effectively.	*0.4	/2
Buttons			Buttons are intuitively designed and strategically placed, facilitating user interaction and navigation within the web-based application.	Button is poorly designed or placed, hindering user interaction and detracting from the overall usability of the web-based application.	*0.4	/2
Tables	Table used is complex but appropriately to present data in a clear and organized manner, enhancing user understanding and accessibility of information.	The table is moderately complex but present. The table enhances user understanding and accessibility of information.	Table may be adequate. It may be needing some improvement for clarity.	The table is too simple, but clear and in organized manner.	*2	/10

Consistency	.		The website maintains a consistent design, including uniform color schemes, typography, spacing, and alignment across all pages. Content is structured logically	The website lacks consistency, with major variations in design, color, and structure, affecting the overall user experience.	*0.4	/2
TOTAL						40

Name:

Appendix

Rubric: Part B

CO2: Demonstrate thinking skills to solve specific problems with web-based applications (C4, PLO2)

SCORE						
Components	5 (Excellent)	4 (Good)	3 (Fair)	2 (Poor)	Weightage	Mark
Website Design / Layouts	The website design and layout demonstrate exceptional creativity, innovation, and functionality. It effectively captures the purpose and essence of the content web-based application, with a visually appealing layout that enhances user engagement and navigation.	The website design and layout are well-executed, showcasing creativity and functionality based on the content. It effectively serves the purpose of the web-based application, with a clear and organized layout that facilitates user interaction and navigation.	The website design and layout are adequate but may lack some creativity or innovation based on content. It adequately serves the purpose of the web-based application, but there may be minor issues with layout organization or user interaction.	The website design and layout are subpar, lacking creativity, innovation, and functionality. It fails to effectively serve the purpose of the web-based application, with a cluttered or confusing layout that hinders user engagement and navigation.	*2	/10
Navigation and User Experience	Demonstrates strategic thinking in designing intuitive navigation that supports user goals. The layout reflects thoughtful problem-solving—users can access all features within 1 click. Responsive design is seamless. Clear use of UI feedback (e.g., hover states, active menu indicators, breadcrumbs). Navigation enhances both usability and user experience across devices. All internal links and buttons function flawlessly.	Applies thoughtful navigation design to solve user access problems. Most features are accessible within 1–2 clicks. Navigation is mostly responsive and consistent. Minor flaws in visual cues or style do not disrupt user experience. Few dead links or unclear paths. Demonstrates good understanding of effective UX practices.	Demonstrates moderate problemsolving; navigation is functional but lacks efficiency. Key features may take more than 3 clicks to reach or are inconsistently placed. Inconsistent layout or styling affects clarity. Some missing cues (e.g., hover effect, breadcrumbs). Several minor usability issues or broken links indicate limited planning.	Shows minimal effort in addressing navigation-related problems. Design is cluttered or confusing, requiring excessive clicks. Navigation lacks logical structure. Broken or missing links are common. User interaction is inconsistent or non-intuitive, reflecting weak application of UX thinking.	*2	/10
Content and Creation	Content and creation are exceptional, with comprehensive and engaging materials that effectively communicate the purpose and value of the web-based application. The content is well-written, informative, and relevant, adding significant value to the overall user experience.	Content and creation are welldeveloped, with engaging materials that effectively communicate the purpose of the web-based application. The content is informative and relevant, contributing to the overall user experience, but there may be minor areas for improvement in clarity or depth.	Content and creation are adequate but may lack depth or engagement. While the content effectively communicates the purpose of the web-based application, there may be issues with clarity or relevance that detract from the overall user experience.	Content and creation are subpar, with minimal effort invested in developing engaging materials. The content may be superficial or irrelevant, failing to effectively communicate the purpose of the web-based application.	*1	/5

Integration of Multimedia	There are at least three (3) multimedia components. Images, videos, and other multimedia components effectively supplement the textual content, engaging users and enriching their interaction with the application.	There are at least two (2) multimedia components. While there may be minor areas for improvement in multimedia selection or integration, they effectively enhance user engagement and comprehension.	There is at least one (1) multimedia component. While present, they may not effectively supplement the textual content or engage users as intended.	There is multimedia. The multimedia component may be sparse or poorly integrated, detracting from the overall user experience and comprehension.	*1	/5
CSS Styling	There are at least five (5) elements of CSS. CSS styling is exceptional, with visually appealing design elements that enhance the overall aesthetic of the web-based application. The styling is consistent, coherent, and effectively complements the content and layout of the application.	There are at least four (4) elements of CSS. CSS styling is well-executed, with design elements that enhance the visual appeal of the web-based application. While effective, there may be minor inconsistencies or areas for improvement in styling coherence or execution.	There are at least three (3) elements of CSS. CSS styling is adequate but may lack polish or coherence. While present, there may be issues with consistency or execution that detract from the overall aesthetic of the web-based application.	There are at least two (2) elements of CSS. Design elements may be inconsistent, disjointed, or poorly executed, detracting from the overall user experience.	*2	/10
TOTAL						40

-END QUESTION -